

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2
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**ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS**

H2 BIOLOGY

9744/01

Paper 1: Multiple Choice

Friday

17th September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice Optical answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit both question paper and multiple choice answer sheet.

This document consists of **21** printed pages

[Turn over

- 1 Lysosomes vary in shape and size, making them difficult to identify.

What describes a lysosome?

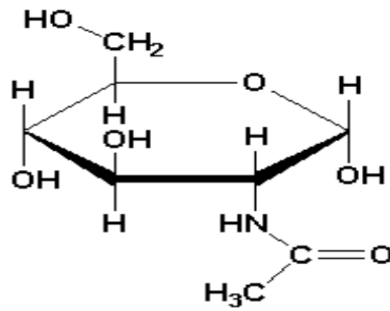
- A a vesicle containing enzymes, enclosed by a double membrane, that is budded off the endoplasmic reticulum
 - B a vesicle containing hydrolytic enzymes and surrounded by a single membrane, found only in phagocytes
 - C a vesicle enclosed by a single membrane, containing several different hydrolytic enzymes that may act inside the cell
 - D a vesicle surrounded by a double membrane, containing enzymes which can hydrolyse damaged organelles in a cell
- 2 The cell surface membranes of some cells are largely made up of phospholipids and cholesterol, with a lower number of transmembrane proteins than usual.

Which transport mechanisms will occur less across these membranes?

- 1 facilitated diffusion
- 2 active transport
- 3 diffusion

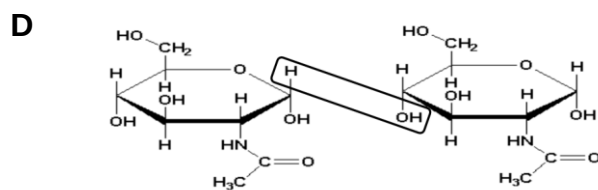
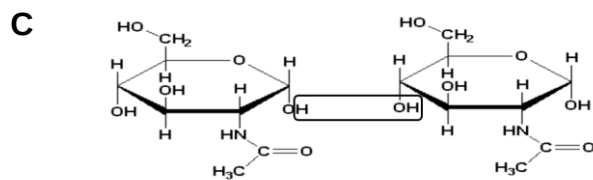
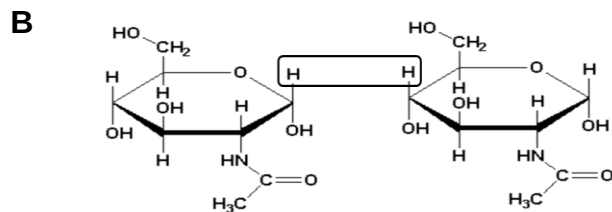
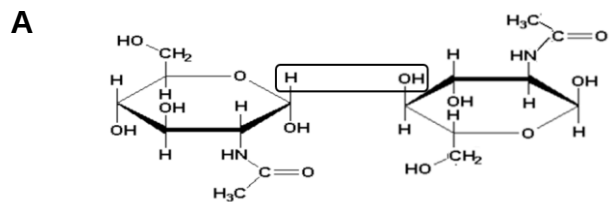
- A 1, 2 and 3
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only

- 3 The diagram shows the structure of a monomer N-acetyl glucosamine.

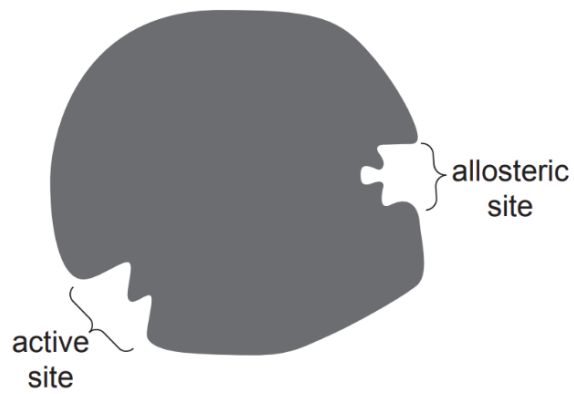


Many N-acetyl glucosamine monomers can be joined by glycosidic bonds to form a polysaccharide called chitin.

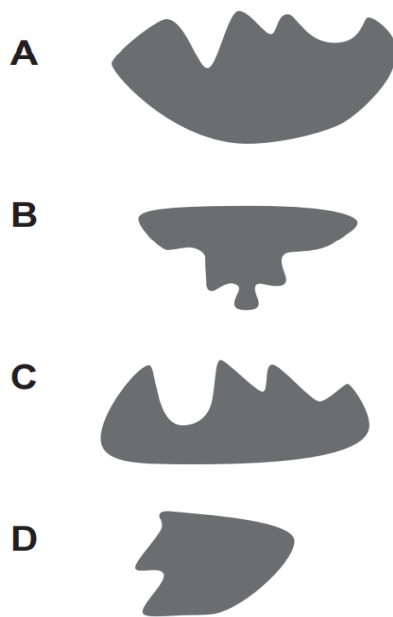
Which diagram shows the formation of a glycosidic bond between two N-acetyl glucosamine monomers?



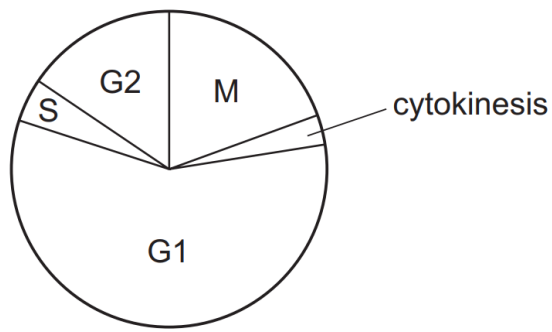
4 The image below shows a diagram of an enzyme.



Which of the following could be a competitive inhibitor of this enzyme?



5 The diagram shows the cell cycle of a mammalian cell.



Cell cycle regulation was done to prevent the cycle from continuing when mistakes are made or when DNA is damaged.

Four of the processes to regulate the cell cycle are described.

- 1 Mitosis is blocked if DNA replication is incomplete.
- 2 Anaphase is blocked if the assembly of chromatids on the spindle is unsuccessful.
- 3 DNA replication is blocked if DNA is damaged.
- 4 DNA replication stops if damage to DNA has not been repaired.

In which phases of the cell cycle do these processes occur?

	Phases			
	1	2	3	4
A	M	G1	S	G2
B	G2	M	G1	S
C	G2	S	G1	M
D	S	G2	M	G1

6 Down's syndrome results from chromosomal aberration of chromosome 21.

This can occur in several ways.

- 1** During gamete formation the homologous pair of chromosomes of chromosome 21 fail to separate. A gamete with two copies of chromosome 21 is fertilised by a gamete with one copy of chromosome 21.
- 2** During early development of the embryo, chromatids of a chromosome 21 fail to separate some cells.
- 3** Translocation of part of chromosome 21 onto chromosome 14 during the development of the embryo. During gamete formation, chromosomes separate normally but may contain an extra chromosome 21.

Some individuals have a form of Down's syndrome called mosaicism. In these cases some of the body cell have the usual two copies of chromosome 21 and other cells have three copies of chromosome 21.

Which of the ways could explain how the mosaicism form of Down's syndrome occurs?

- A** 1 and 2
- B** 1 and 3
- C** 2 only
- D** 3 only

7 Hox genes contain a homeobox sequence of 180 base pairs. Two species have a homeobox sequence of 180 base pairs where 1.7% of the base pairs are different.

Which of the following shows the possible number of amino acids coded for that would be different in the two species?

- A** minimum 0 and maximum 2
- B** minimum 0 and maximum 3
- C** minimum 1 and maximum 2
- D** minimum 1 and maximum 3

- 8** The diagram shows part of the DNA sequence of a gene and a mutated sequence of the same gene.

normal DNA sequence ...CCG GAT TAT TGC GAG AAA TGG CAT TCT AGG...

mutated DNA sequence ...CCG GAT GTA TTG CGA GAA ATG CAT TCT AGG...

What are possible effects of the mutated sequence?

- 1** the presence of mRNA stop codons, UAG, UAA or UGA
- 2** a change in the sequence of amino acids
- 3** a non-functional protein
- 4** ribosomes cannot translate the mRNA

- A** 1, 2 and 3
B 1, 3 and 4
C 1 and 4 only
D 2 and 3 only

- 9** Which features are typical of eukaryotic genomes?

- 1** Genes widely separated apart
- 2** Code almost entirely for proteins
- 3** Include repetitive sequences
- 4** Simultaneous transcription of related genes

- A** 1 and 2
B 1 and 3
C 2 and 3 only
D 3 and 4 only

10 Which of the following are DNA sequences?

- 1** Promoter
- 2** Enhancer
- 3** Silencer
- 4** Repressor
- 5** Activator
- 6** Operator

- A** 1, 2 and 3
B 1, 2 and 6
C 4, 5 and 6
D 1, 2, 3 and 6

11 Most eukaryotic mRNA that is translated has a poly(A) tail at the 3' end of each molecule.

- *In vitro* investigations show that shortening an mRNA's poly (A) tail inhibits translation.
- Studies show the presence in early embryos of stored maternal mRNA. These stored maternal mRNA in early embryos were transcribed from DNA during egg development and then polyadenylated with short poly(A) tails.
- When these mRNA are needed in embryonic development, their poly(A) tails are extended.

What may be concluded from these findings?

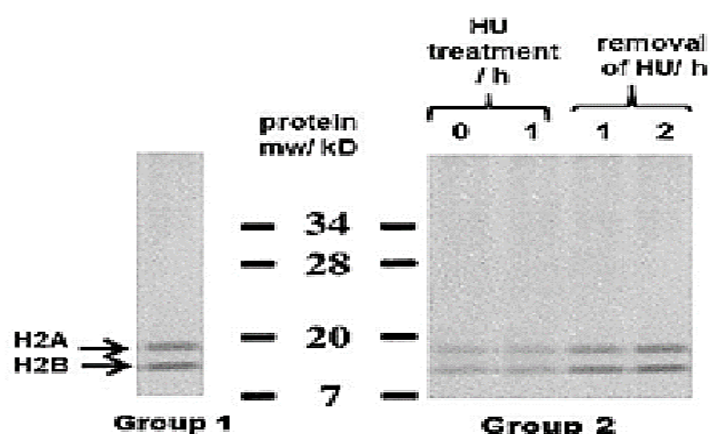
- 1** Adding a poly(A) tail is essential to stabilise mRNA and protect it from being broken down.
- 2** The poly(A) tail is not encoded by the DNA of the transcribed gene.
- 3** A long poly(A) tail acts as a switch to trigger translation.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

- 12** A scientist investigated the mode of action of a drug, hydroxyurea (HU), that was known to prevent cell cycle progression in the parasite, *Leishmania*.

In the experiment, two groups of *Leishmania* parasites are used. **Group 1** is the untreated control. **Group 2** is incubated in a culture medium with HU for 1 hour before being transferred to a fresh medium without HU.

The effect of HU on histone synthesis was investigated by incubating parasite cells in a mixture of amino acids containing methionine that has been labelled with radioactive isotope sulfur. Histone synthesis was measured by the intensity of the dark bands shown in the autoradiograph. The figure below is an autoradiograph showing the levels of H2 histone proteins produced by Group 1 and Group 2 in the experiment.



With reference to the results above, which of the following **cannot** be a possible mode of action of HU?

- A** HU prevents the formation of the Transcription Initiation Complex.
- B** HU stops the binding of Translation Initiation factors to the small ribosomal subunit.
- C** HU enhances the action of poly(A) polymerase.
- D** HU promotes the addition of ubiquitin to histone proteins.

- 13** In a human cell with a normal tumour suppressor gene, *p53*, if the cell is damaged, *p53* is expressed and protein p53 is synthesised. Protein p53 causes the cell to undergo apoptosis when necessary. Some types of cancer are caused by the mutation of tumour suppressor gene *p53*, preventing the synthesis of protein p53.

Normal adenoviruses produce a viral protein that inhibits protein p53, disrupting the cell's normal response to damage. When there is no functional protein p53, then the virus can replicate and ultimately kill the cell.

Recently, scientists discovered a mutant adenovirus that synthesises a non-functional version of the viral protein, which is unable to inactivate protein p53.

Which statement correctly describes the outcome of cells from a cancer patient when infected by the mutant adenovirus?

- A** This mutant virus would only cause the death of normal cells.
- B** This mutant virus would only cause the death of tumour cells.
- C** This mutant virus would cause death of both normal cells and cancer cells.
- D** This mutant virus would not cause the death of any cell.

- 14** What is true about second messengers in a cell signalling system?

- 1** They can be linked to a protein kinase cascade to amplify the signal.
- 2** They are activated only when they bind to extracellular signalling molecules.
- 3** They always cross cell surface membranes to initiate cellular processes.
- 4** Membrane-bound receptors often trigger their release or synthesis.

- A** 1, 2 and 3
- B** 1, 3 and 4
- C** 1 and 4 only
- D** 2, 3 and 4

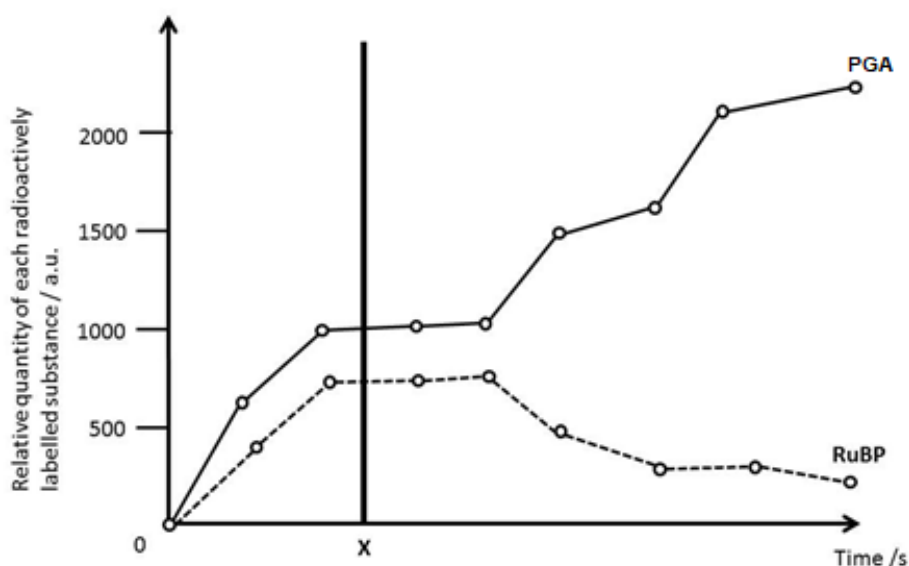
- 15** Which of the following statement about anaerobic respiration is correct?

- A** Animal cells produce pyruvate from lactate, so as to regenerate NAD⁺.
- B** Plant cells use their chloroplasts to synthesise ATP for cellular activities when aerobic respiration in mitochondria cannot occur.
- C** From one molecule of glucose, aerobic respiration yields 10 to 15 times more ATP than anaerobic respiration.
- D** From one molecule of glucose, anaerobic respiration in both plant and animal cells will yield the same amount of ATP.

- 16** Many chemicals are used as herbicides as they are able to inhibit photosynthesis in plants. Diquat, a herbicide, inhibits Photosystem I by intercepting electrons from an electron carrier.

To investigate the effect of Diquat on light-independent reactions, an illuminated suspension of photosynthesizing algae *Chlorella* was treated with carbon dioxide containing a heavy isotope of carbon, ^{14}C . The levels of ribulose biphosphate (RuBP) and glycerate-3-phosphate (PGA) were subsequently measured.

The figure shows the results of the experiment. Point X indicates the point at which Diquat was added to the *Chlorella* suspension.



Which row correctly explains why the level of PGA increases and RuBP decreases after point X?

	<i>Carbon fixation</i>	<i>PGA reduction</i>	<i>RuBP regeneration</i>
A	✓	✓	X
B	X	X	✓
C	✓	X	X
D	✓	X	✓

17 Which statement uses genetic terms correctly?

- A** For a gene with multiple alleles, whether each allele is dominant or recessive is independent of the other alleles.
- B** For a gene with two alleles showing incomplete dominance, the genotypic ratio in a cross involving two heterozygotes is always the same as the phenotypic ratio.
- C** Homologous chromosomes have the same alleles at the same loci.
- D** X-linked recessive diseases only occur in males since females can only be carriers.

18 In humans, a certain disease is caused by inheriting two recessive alleles of gene A/a.

Identical twins occur once in approximately 300 births.

A woman who is heterozygous for gene A/a is about to give birth. The father is also heterozygous for this gene.

What is the probability that the woman will give birth to identical twins that do **not** have the disease?

- A** approximately 1 in 300
- B** approximately 1 in 400
- C** approximately 1 in 800
- D** approximately 1 in 1200

19 An unspotted black female dog was crossed with a spotted black male dog. She gave birth to six puppies of four different phenotypes: unspotted black, unspotted brown, spotted black and spotted brown. Some of the puppies were adopted and live in different places. As the puppies got older, these phenotypes did not change.

Some of the phenotypes of the puppies were different from the phenotypes of either of their parents.

What is needed to explain the appearance of these new phenotypes?

- A** Codominance and meiosis
- B** Multiple alleles and random fertilisation
- C** Environmental factors and random fertilisation
- D** Random fertilisation and meiosis

- 20** A scientist starts her experiment with an ampicillin-resistant strain of bacteria which has a non-functional *lac Z* gene. She adds plasmids which contain a functional copy of *lac Z* gene to a tube of bacteria (+). A control tube (-) with the same bacteria but no plasmid was prepared.

Both tubes are incubated under the appropriate conditions for growth and plasmid uptake via transformation. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.

Plate A: glucose medium

Plate B: glucose medium with ampicillin

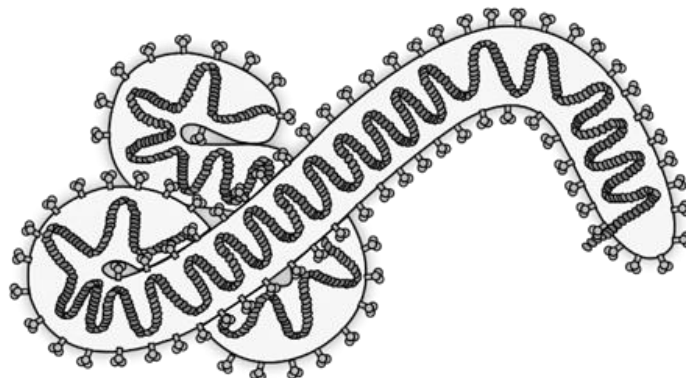
Plate C: lactose medium with ampicillin

Which row correctly identifies the plates with growth (✓) or no growth (✗)?

	Bacterial strain with added plasmid (+)			Bacterial strain with no plasmid (-)		
	Plate A	Plate B	Plate C	Plate A	Plate B	Plate C
A	✓	✓	✓	✓	✓	✓
B	✓	✓	✓	✓	✓	✗
C	✓	✓	✓	✓	✗	✗
D	✓	✓	✗	✓	✓	✗

- 21** Ebola viruses are RNA viruses endemic to regions of west and equatorial Africa. They are pathogens that are primarily transmitted by human-to-human contact with infected body fluids and result in high mortality.

The diagram shows the structure of the Ebola virus.

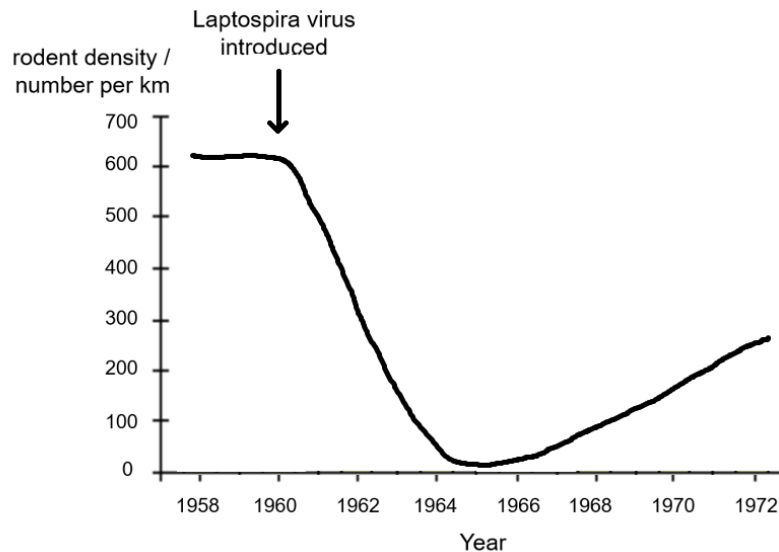


The RNA genome of Ebola viruses cannot be directly translated to synthesise viral proteins.

Which row correctly describes Ebola viruses?

	presence of envelope	genome	use of host RNA polymerase
A	absent	positive-sense RNA	no
B	absent	negative-sense RNA	yes
C	present	positive-sense RNA	yes
D	present	negative-sense RNA	no

- 22** The graph shows the rodent density in a region of Italy from 1958 to 1972. The Laptospira virus, which causes internal bleeding, was introduced in Italy in 1960. The Laptospira virus is estimated to have infected 95% of the rodent population in 1964.



Using the information given, which of the following is a plausible explanation for the change in rodent density?

- A** Random mutations to develop resistance against the virus occur in rodents in response to Laptospira virus infection.
 - B** The rodent population have developed resistance to Laptospira virus since 1965.
 - C** Antigenic shift occurred in Laptovirus population in 1964.
 - D** Genetic drift occurred in 1962 resulting in the drastic decline in rodent density.
- 23** In a particular cycle of a polymerase chain reaction (PCR), DNA is denatured in high temperatures, before the annealing stage in 55°C.

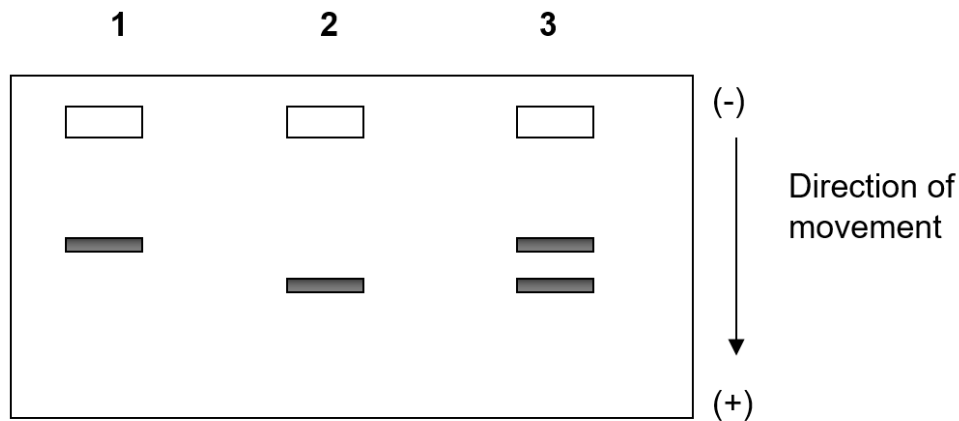
What could be a possible reason for why majority of the denatured DNA strands re-anneal with themselves instead of annealing to primers?

- A** The primers are in excess.
- B** The primers anneal only to the 3' end of the denatured DNA.
- C** The denatured DNA concentration exceeds the primer concentration.
- D** The temperature is too low for primers to anneal to the denature DNA strands.

- 24** The β -globin gene can exist in two different alleles termed HbA (the normal allele) and HbS (the allele that causes sickle cell anaemia in homozygotes).

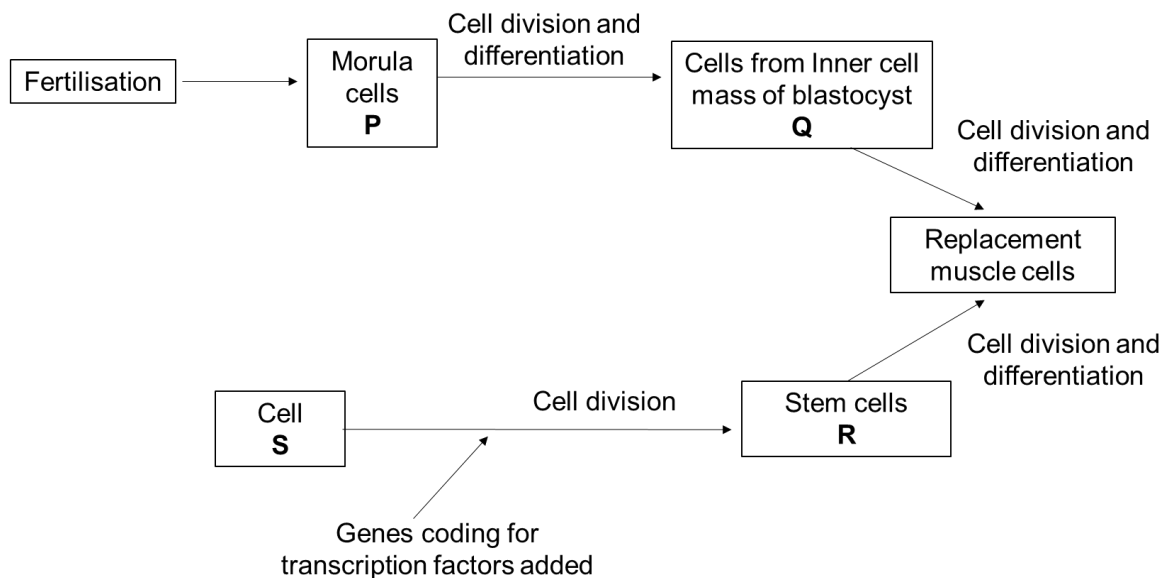
The polypeptides that are coded for by these two alleles differ by one amino acid. Blood samples from 3 individuals were obtained and the **proteins** were separated by gel electrophoresis.

Based on the gel electrophoresis pattern below, which of the following are **false**?



- A** Genotype of individual 1 is Hb^S Hb^S because valine in mutant haemoglobin that causes sickle cell anaemia is neutral / uncharged.
- B** Genotype of individual 2 is Hb^A Hb^A because glutamic acid in the normal haemoglobin is positively charged.
- C** Genotype of individual 3 is Hb^A Hb^S because both alleles are expressed.
- D** In malaria-infected countries, people with the same genotype as individual 3 are at a selective advantage.

25 The diagram shows two ways in which replacement muscle cells can be produced.



Which row correctly describes the cells P, Q, R and S?

	P	Q	R	S
A	pluripotent	multipotent	totipotent	muscle cell
B	pluripotent	totipotent	induced pluripotent	egg cell
C	totipotent	pluripotent	induced pluripotent	skin cell
D	totipotent	pluripotent	totipotent	muscle stem cell

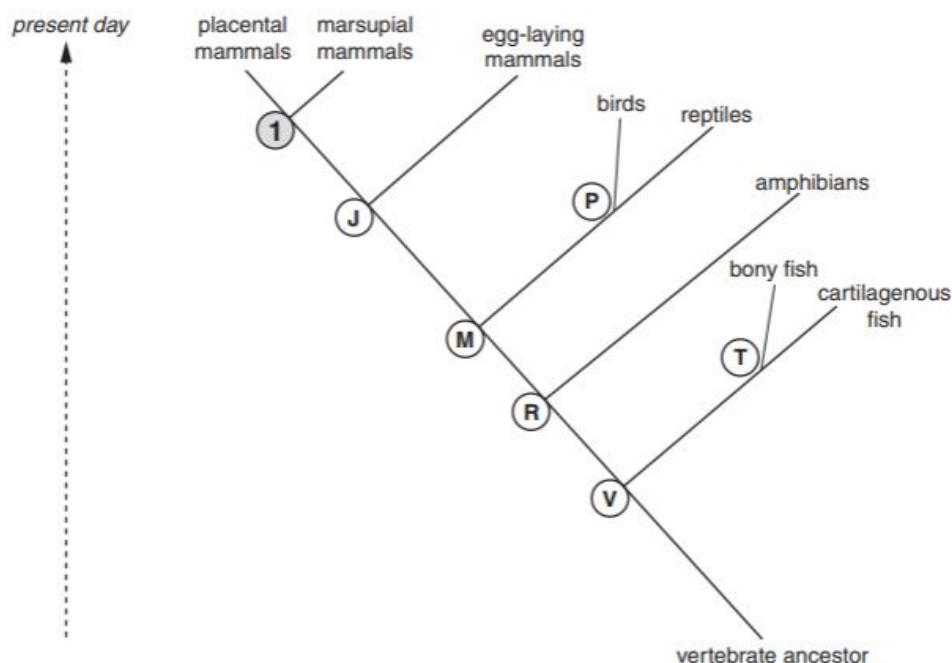
26 In which of the following situations would evolution be slowest for an interbreeding population?

	Migration	Selection pressure	Variation due to mutation
A	Absent	Low	Low
B	Absent	High	High
C	High	Low	High
D	High	High	Low

- 27** Cytochrome c is a protein found in living organisms. The structure of cytochrome c varies between different organisms. However, closely related organisms have similar cytochrome c.

The figure below shows a possible evolutionary tree for vertebrates.

Common ancestors are indicated by the number 1 and various letters.



Which of the following are correct deductions based on the phylogenetic tree?

The cytochrome c of:

- 1** Common ancestors P and M are equally similar in structure to common ancestor J.
 - 2** Common ancestor T is less similar in structure to the vertebrate ancestor than common ancestor P.
 - 3** Common ancestor J is most similar in structure to common ancestor 1.
 - 4** Placental mammals and marsupial mammals share more similarities with each other than bony fish and cartilaginous fish.
- A** 3 only
- B** 1 and 3 only
- C** 2 and 4 only
- D** 3 and 4 only

- 28** Soay sheep, a primitive breed of domesticated sheep, live wild on the small and remote Scottish island of Hirta. Studies have shown that the mean size of an adult Soay sheep has been decreasing over the past thirty years.

Which statements could explain the decrease in mean size of adult Soay sheep on Hirta?

- 1** Stabilising selection is occurring, with the largest and smallest sheep being selected against.
- 2** Small lambs are less likely to survive their first winter than large lambs.
- 3** Weather change has led to milder winters, so smaller lambs are surviving to adulthood.
- 4** Food has become scarcer and smaller sheep need less food than larger sheep.

- A** 1 and 3
- B** 1 only
- C** 2 and 4
- D** 3 and 4

- 29** Smallpox is a contagious, airborne disease caused by the variola virus. The variola virus has a DNA genome and has a low mutation rate.

The figure below shows the number of cases of smallpox per year in country X.



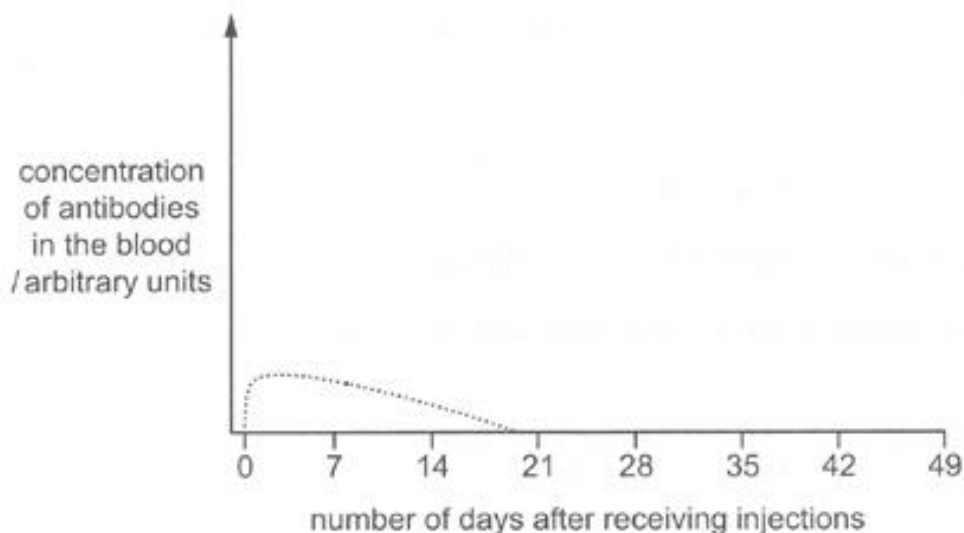
Which statements could be inferred from these data?

- 1** The gradual fall in the number of reported cases of smallpox in the 1960s could be due the development of antibiotics.
- 2** The gradual fall in the number of reported cases of smallpox in the 1960s and zero cases reported in 1975 could be due to a vaccination programme against smallpox.
- 3** The surge in the number of cases in 1974 could be due to the virus having a noticeably high mutations deliberated at its genome.
- 4** The surge in the number of cases in 1974 could be due to insufficient proportion of people being vaccinated to break the disease transmission cycle.

- A** 1 and 3 only
B 2 and 4 only
C 2, 3 and 4 only
D 1, 2, 3 and 4

30 A patient sent to the hospital received one injection as treatment.

The graph shows how the injection affected the concentration of antibodies in the blood.



Which of the following is correct?

- A** Through this treatment, B lymphocytes of the patient is activated to plasma cells which secrete antibodies into the blood.
- B** The patient received passive immunity through the treatment.
- C** The patient will be protected by the immune system against a re-infection by the same pathogen due to generation of memory cells.
- D** The drop of antibody concentration to zero at day 19 could be due to the death of plasma cells which secrete antibodies